

NATIONAL INSTITUTE OF TECHNOLOGY PATNA

PATNA, BIHAR

MID-TERM EXAMINATION-2023

COURSE: B.TECH.

BRANCH: ECE

SEM: II

FUNDAMENTALS OF COMMUNICATION ENGINEERING (EC25102)

DATE: 17-05-2023

TIME: 2 HOURS

MAXIMUM MARKS: 30

Note: Attempt any FIVE questions.

- ① Draw the block diagram of analog communication. Explain in detail the working of each and every block. [6] (CO1)
- ② Explain the concept of band-limitation in the communication engineering. Why the modulation is needed for distant communication? Why sinusoidal waveform is preferable as carrier in the modulation? [6] (CO1)
- ③ A carrier signal $10\cos 8\pi \times 10^5 t$ is amplitude modulated by a message signal $6\sin \pi \times 10^4 t$. Let, the antenna resistance is 1Ω . [6] (CO4)
a) Find the total power after modulation and the modulation efficiency
b) Plot the spectrum of the modulated signal
- ④ A given message signal has two frequency components. The modulation index corresponding to these two frequency components are μ_1 and μ_2 . If a carrier (say, $A_c \cos 2\pi f_c t$) is modulated by the given message signal, prove that the total modulation index (μ_t) will be: $\mu_t = \sqrt{\mu_1^2 + \mu_2^2}$ and overall modulation efficiency (η) will be: $\eta = \frac{\mu_t^2}{\mu_t^2 + 2}$ [6] (CO4)
- ⑤ Explain in detail the envelop detector of Double-Side-Band AM modulation with the help of circuit diagram and waveforms. How the discharging time constant of capacitor depends on the carrier frequency and message frequency? [6] (CO4)
- ⑥ Compare among DSB, DSB-SC and SSB-SC in tabular format in terms of equations, frequency spectrum, generation, detection, power, and bandwidth. [6] (CO4)
- ⑦ Explain in detail; why SSB-SC modulation cannot be used for complete audio signal and video signal? [6] (CO4)

NATIONAL INSTITUTE OF TECHNOLOGY PATNA

PATNA, BIHAR

END-TERM EXAMINATION-2023

COURSE: B.TECH.

BRANCH: ECE

SEM: II

FUNDAMENTALS OF COMMUNICATION ENGINEERING (EC25102)

DATE: 18-06-2023

TIME: 3 HOURS

MAXIMUM MARKS: 60

Note: Attempt any **SIX** questions.

1. Explain in detail the phase discrimination method to generate single side band suppressed carrier (SSB-SC) signal with the help of block diagram and mathematical expressions. Why this method is not preferable for the generation of SSB-SC? [10] (CO4)

2. Analyse the WBFM with the help of general mathematical expressions, frequency spectrum, power, practical bandwidth, and efficiency. [10] (CO4)

3. Explain in detail the phase locked loop (PLL) method to detect the frequency modulated signal with the help of block diagram and mathematical expressions. [10] (CO4)

4. a) An FM signal is given by:

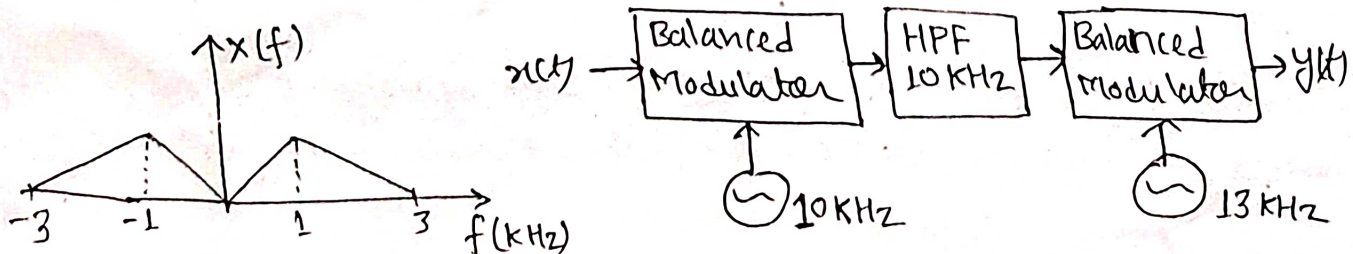
$$s(t) = 10 \cos(2\pi \times 10^6 t + 0.2 \sin(2\pi \times 10^3 t))$$

It is passed through cascaded frequency multiplier of having multiplying constants of 4 and 5, respectively. Find bandwidth and transmitted power at the output of each of the multiplier.

[5] (CO4)

b) Consider a system shown in the following Fig. 1. Let $X(f)$ and $Y(f)$ denote the Fourier transformations of $x(t)$ and $y(t)$, respectively. The ideal high pass filter (HPF) has the cut-off frequency 10 kHz. Find out the positive frequencies corresponding to spectral peaks of $Y(f)$.

[5] (CO4)



5. a) The signal $m(t)$ as shown in Fig. 2 is applied both to a phase modulator and a frequency modulator having the same carrier frequency. Find out the ratio of phase sensitivity of phase modulator to the frequency sensitivity of the frequency modulator for the same maximum phase deviation. [5] (CO4)

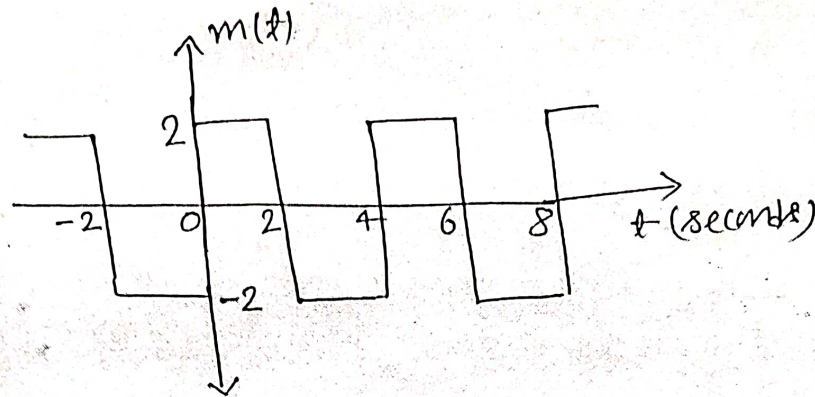


Fig.2 Figure for Q.5(a)

- b) A message signal of $8 \sin(2\pi \times 10^3 t)$ is transmitted through PCM system. Sampling rate is 50% higher than Nyquist rate. Minimum signal to quantization noise ratio (SQNR) should be 22dB. Find out transmission bandwidth and SQNR. [5] (CO3, CO4)
6. a) Describe in detail the line coding techniques with the help of waveforms for the binary data 01101001. Also discuss merits and demerits of each of the line coding techniques. [5] (CO3, CO4)
- b) Differentiate between step-index fiber and graded index fiber in tabular format by considering the parameters: refractive index profile, numerical aperture, ray propagation in the core, dispersion, and modes. [5] (CO3, CO4)
7. Explain in detail the followings:
- a) Frequency reuse concept in cellular wireless communication. [5] (CO5)
 - b) Orbital mechanics of satellite communications [5] (CO5)
8. Explain in detail the followings:
- a) Dispersion mechanism in optical fiber [5] (CO5)
 - b) Optical sources: LED and LASER [5] (CO5)
9. Explain in detail the followings:
- a) Multiple access techniques in wireless communication [5] (CO5)
 - b) Internet of the things (IOT) and its application [5] (CO7)